

VLADIVOSTOK INTL, Russian Federation

WMO: 319741

Lat: 43.399N

Lon: 132.148E

Elev: 46

StdP: 14.67

Time Zone: 10.00 (E10)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.9	-10.7	-24.0	1.5	-9.7	-20.5	1.8	-6.3	21.0	12.4	19.3	11.9	1.6	10	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.6	85.6	72.5	82.3	71.2	79.1	69.3	75.5	81.4	73.8	79.0	72.1	76.9	7.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.6	125.1	78.6	71.8	117.9	76.7	70.1	111.0	74.8	39.0	80.5	37.3	78.6	35.9	76.1	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.3	17.2	15.4	DB	-21.8	91.0	4.7	2.5	-25.1	92.8	-27.8	94.3	-30.5	95.7	-33.8	97.4
			WB	-21.9	78.0	4.6	1.8	-25.2	79.3	-27.9	80.3	-30.5	81.3	-33.8	82.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	7.3	13.6	28.3	42.7	54.1	62.0	69.5	71.2	61.1	46.3	28.2	11.3
	DBStd	23.27	7.80	8.99	8.77	5.93	5.30	4.84	4.87	4.71	5.98	7.60	10.42	8.74
	HDD50	5289	1323	1018	673	229	22	0	0	0	3	165	656	1200
	HDD65	9001	1788	1438	1138	668	340	114	15	7	142	581	1105	1665
	CDD50	2168	0	0	0	11	147	360	605	659	337	49	0	0
	CDD65	406	0	0	0	0	0	24	155	201	26	0	0	0
	CDH74	2347	0	0	0	2	58	200	803	1131	152	1	0	0
	CDH80	526	0	0	0	0	4	40	213	257	12	0	0	0
Wind	WSAvg	6.3	6.3	6.6	7.2	7.4	7.0	6.5	5.8	5.3	5.4	6.1	6.1	6.2
Precipitation	PrecAvg	29.3	0.6	0.7	1.1	1.5	3.1	3.6	5.5	5.4	3.0	2.2	1.6	1.0
	PrecMax	38.7	3.3	2.2	4.1	3.4	5.7	7.2	10.7	13.3	9.0	6.1	4.1	3.1
	PrecMin	18.9	0.0	0.0	0.0	0.2	0.5	1.7	1.6	0.8	0.5	0.5	0.3	0.1
	PrecStd	5.3	0.8	0.6	0.8	0.8	1.3	1.3	2.3	2.7	2.1	1.2	1.0	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	42.1	58.9	70.6	79.1	84.5	89.7	89.4	81.0	71.5	58.9	42.6
		MCWB	30.5	34.1	43.8	51.3	58.3	68.3	75.5	73.9	67.8	57.9	50.2	39.9
	2%	DB	30.4	37.8	50.3	64.6	75.1	79.1	85.2	85.8	78.0	67.7	53.7	35.8
		MCWB	26.3	33.3	38.6	47.9	57.9	65.6	72.8	72.9	66.1	57.0	45.9	31.3
	5%	DB	26.6	35.2	46.3	60.5	71.2	75.2	82.2	82.6	75.3	64.4	48.4	30.6
		MCWB	22.7	31.1	36.5	45.9	55.8	64.5	71.6	71.8	64.5	54.8	42.4	26.9
	10%	DB	22.6	31.2	42.0	55.6	66.4	71.3	78.6	80.3	72.0	61.0	44.3	26.8
		MCWB	19.0	26.7	34.1	43.5	54.4	63.1	70.3	70.6	62.7	53.2	39.5	23.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.5	37.2	44.6	53.0	62.2	71.7	78.0	77.6	72.1	62.9	52.0	40.1
		MCDB	33.8	39.3	55.4	67.6	75.0	82.1	85.4	83.7	75.5	67.0	56.2	41.5
	2%	WB	26.2	34.3	40.3	49.8	59.5	68.2	75.3	75.9	69.2	60.0	47.6	32.2
		MCDB	29.3	37.0	47.9	60.4	70.1	75.3	82.5	81.5	73.7	65.0	51.9	34.2
	5%	WB	22.2	30.4	37.4	47.3	57.6	65.7	73.2	74.6	67.3	57.4	43.4	27.4
		MCDB	26.1	34.3	44.0	56.1	67.1	72.5	78.7	79.6	72.7	62.8	47.1	30.5
	10%	WB	18.5	26.4	35.0	45.0	55.7	63.9	71.5	73.1	65.3	54.6	39.1	23.2
		MCDB	22.0	30.8	40.7	53.0	64.2	69.9	76.3	77.6	70.7	59.5	43.4	26.7
Mean Daily Temperature Range	5% DB	MDBR	23.2	24.4	21.5	22.2	20.9	15.5	14.3	15.6	21.1	23.6	19.8	21.2
		MCDBR	28.6	26.3	27.5	31.3	31.4	25.1	22.0	19.8	23.2	26.4	25.5	25.5
		MCWBR	24.8	21.0	16.9	17.0	15.5	13.5	11.1	9.3	12.5	16.4	18.3	22.1
	5% WB	MCDBR	26.9	22.5	24.1	26.7	25.2	20.3	17.5	14.9	17.9	22.0	22.8	24.7
		MCWBR	24.0	19.3	16.0	15.6	13.6	11.6	9.5	7.9	10.5	15.9	18.2	22.3
Clear-Sky Solar Irradiance	taub	0.328	0.388	0.473	0.573	0.592	0.567	0.534	0.487	0.436	0.448	0.395	0.345	
	taud	2.128	1.957	1.828	1.690	1.716	1.861	2.013	2.145	2.200	2.021	2.065	2.091	
	Ebn at Noon	244	244	237	224	225	232	238	246	249	222	213	221	
	Edh at Noon	32	45	58	73	73	63	54	46	40	42	34	30	
All-Sky Solar Radiation	RadAvg	699	1009	1331	1472	1594	1619	1457	1428	1331	989	689	574	
	RadStd	41	61	85	136	144	213	191	141	117	76	56	36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page